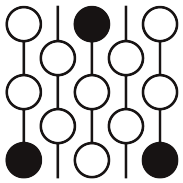
Review of Problem 5 – Siege



First Proof Third Batch
Reviewer 1
October 8–9, 2026

1 Review

The solution is mathematically correct. It correctly identifies that the answer is no and construct a counterexample using the construction of infinite cyclic projective plane. This construction of hyperfield already appears in the literature, for example in the work on Connes–Consani. The new observation is this construction can give a negative answer to the question. This construction is essentially identical to other AI system submissions and notably different from the human solution. Some attributions are missing and the exposition can be improved. Some proofs are reproduced from the literature and could have been omitted, with a proper citation.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
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Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
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Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
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Score: 0

2 Submission with inline comments

3 The problem

Question 1. Let $\mathbb{H}$ be a hyperfield with multivalued addition operator $\boxplus$. Let $x, y, z \in \mathbb{H}$ be elements. Suppose that $0 \in x \boxplus y \boxplus z$. Does it then follow that

$$0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus -xyz \boxplus -xyz \boxplus -xyz\,?$$

4 The result

The answer is *no*.

Theorem 1. *There is a hyperfield $\mathbb{H}$ with elements x, y, z such that*

$$0 \in x \boxplus y \boxplus z \qquad \textit{but} \qquad 0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

One may take $\mathbb{H}$ infinite, with multiplicative group $\mathbb{H}^\times = \langle t \rangle \cong \mathbb{Z}$ infinite cyclic, and $x = t$, $y = t^4$, $z = 1$.

Conventions. "Hyperfield" is meant in the standard sense of Krasner [4, 5]; see also [6, 1]. Thus a hyperfield is a set $\mathbb{H}$ with a distinguished element 0 and the following structure:

- a commutative, associative hyperoperation $\boxplus$ with nonempty values, for which 0 is neutral;
- unique negatives: for each a there is exactly one b, denoted $-a$, with $0 \in a \boxplus b$;
- reversibility: $c \in a \boxplus b \Rightarrow a \in c \boxplus (-b)$;
- a multiplication making $\mathbb{H} \setminus \{0\}$ an abelian group, with $0 \cdot h = h \cdot 0 = 0$ and $w(a \boxplus b) = wa \boxplus wb$.

Sums of sets are taken elementwise, $A \boxplus B = \bigcup_{a \in A,\, b \in B} a \boxplus b$, and $a \boxplus B := \{a\} \boxplus B$. By associativity, iterated sums do not depend on the bracketing (nor, by commutativity, on the order of the summands).

5 Overview of the proof

When $\mathbb{H}$ is a quotient $F/K^\times$ of a field, the answer is yes, because of the identity $X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$ (Remark 8). A counterexample must therefore come from a hyperfield that is not of this kind, and we use hyperfields built from projective planes.

Given an abelian group G acting regularly on the points of a projective plane by collineations, with at least five points on each line, set $\mathbb{H} = G \cup \{0\}$, $a \boxplus a = \{0, a\}$ and $a \boxplus b = \mathrm{L}(a, b) \setminus \{a, b\}$ for $a \neq b$, where $\mathrm{L}(a, b)$ is the line through a and b. This is a hyperfield in which every element is its own negative (Proposition 3); a construction of this kind appears in the work of Connes and Consani [2]. In such an $\mathbb{H}$, with $z = 1$, the hypothesis $0 \in x \boxplus y \boxplus 1$ holds as soon as $1, x, y$ are distinct and collinear, while the six-term sum equals $(\mathcal{M} \cup \{0\}) \boxplus 1$, where $\mathcal{M}$ is the line through x^3, y^3, xy (if these are distinct). This sum avoids 0 whenever $1 \notin \mathcal{M}$ (Lemma 4). So we need a plane in which $1, x, y$ are collinear but 1 is not on the line through x^3, y^3, xy; by the remark above, no quotient of a field can serve here.

Such a plane is obtained on the point set $\mathbb{Z}$ (written additively, so that n stands for t^n) from an infinite perfect difference set $D \subset \mathbb{Z}$, whose translates $g + D$ serve as lines (Lemma 6). A greedy construction (Lemma 7), in the spirit of M. Hall's infinite cyclic planes [3], produces such a D

containing the seed $\{0, 1, 4, 103, 105, 112\}$ and not containing 100. Then $0, 1, 4$ lie on the line D, the points $3, 12, 5$ (corresponding to x^3, y^3, xy) lie on the line $-100 + D$, and 0 does not.

All arguments below are given in full; the only external input is the definition of a hyperfield.

6 Hyperfields from projective planes with a regular abelian group

Let G be an abelian group, written multiplicatively. Consider an incidence structure with point set G and a family of subsets of G, called *lines*, satisfying:
(P1) any two distinct points lie on exactly one line;
(P2) any two distinct lines meet;
(P3) every line has at least 5 points;
(P4) for every $w \in G$, multiplication by w maps lines to lines.
By (P1) two distinct lines have at most one common point, and by (P2) they have at least one; so two distinct lines meet in exactly one point. For distinct points a, b we write $\mathrm{L}(a, b)$ for the line through a and b.

Definition 2. Put $\mathbb{H} = G \cup \{0\}$. Let 0 be neutral: $0 \boxplus a = a \boxplus 0 = \{a\}$ for all $a \in \mathbb{H}$ (so $0 \boxplus 0 = \{0\}$). For $a, b \in G$ set

$$a \boxplus a = \{0, a\}, \qquad a \boxplus b = \mathrm{L}(a, b) \setminus \{a, b\} \quad (a \neq b).$$

Multiplication is the group law on G, extended by $0 \cdot h = h \cdot 0 = 0$.

Proposition 3. *$\mathbb{H}$ is a hyperfield, and $-a = a$ for every $a \in \mathbb{H}$.*

> **Reviewer 1**
>
> This construction appears in [2, Example 4.7] which is based on "A Theorem in Finite Projective Geometry and Some Applications to Number Theory" by J. Singer in 1938. While [2] was mentioned in the introduction, it should be mentioned explicitly here. It would have sufficed to provide a citation rather than reproducing the same proof.

Proof. Commutativity and nonempty values. Commutativity is clear. For $a \neq b$ in G, the set $a \boxplus b$ has at least 3 points by (P3), so every value is nonempty.

Negatives. Let $a \neq 0$. For $b \in G$ with $b \neq a$ we have $a \boxplus b \subseteq G$, so $0 \notin a \boxplus b$. Also $0 \boxplus a = \{a\}$ does not contain 0, while $0 \in a \boxplus a$. Hence a is the unique negative of a. The element 0 is its own unique negative.

Reversibility. Since $-b = b$, we must show $c \in a \boxplus b \Rightarrow a \in c \boxplus b$.

- If $a = 0$, then $c = b$ and $0 \in b \boxplus b$; this includes the case $b = 0$.
- If $b = 0$, then $c = a$ and $a \in a \boxplus 0$.
- If $a = b \neq 0$, then $c \in \{0, a\}$. If $c = 0$, then $a \in 0 \boxplus a$. If $c = a$, then $a \in a \boxplus a$.
- If $a \neq b$, both nonzero, then $c \in \mathrm{L}(a, b) \setminus \{a, b\}$. Since $c \neq b$, $\mathrm{L}(c, b) = \mathrm{L}(a, b)$, so $c \boxplus b = \mathrm{L}(a, b) \setminus \{c, b\}$. This set contains a, because $a \neq c$ and $a \neq b$.

Distributivity. The case $w = 0$ is trivial, so let $w \in G$. Multiplication by w is a bijection of G that maps lines to lines by (P4). Hence $w\,\mathrm{L}(a, b)$ is a line through wa and wb, i.e. $w\,\mathrm{L}(a, b) = \mathrm{L}(wa, wb)$, and so $w(\mathrm{L}(a, b) \setminus \{a, b\}) = \mathrm{L}(wa, wb) \setminus \{wa, wb\}$. Also $w\{0, a\} = \{0, wa\} = wa \boxplus wa$, and the cases involving 0 are trivial.

Associativity. If one of a, b, c is 0, both $(a \boxplus b) \boxplus c$ and $a \boxplus (b \boxplus c)$ equal the sum of the other two, because 0 is neutral.

Now let a, b, c be nonzero and put $S(a, b, c) := (a \boxplus b) \boxplus c$. We show that S depends only on the multiset $\{a, b, c\}$. Then, using commutativity,

$$a \boxplus (b \boxplus c) = (b \boxplus c) \boxplus a = S(b, c, a) = S(a, b, c) = (a \boxplus b) \boxplus c.$$

Case 1. $a = b = c$. Then $S = (0 \boxplus a) \cup (a \boxplus a) = \{0, a\}$.

Case 2. The multiset is $\{a, a, c\}$ with $a \neq c$; let $L = \mathrm{L}(a, c)$. Up to commutativity there are two orderings.

- $(a \boxplus a) \boxplus c = (0 \boxplus c) \cup (a \boxplus c) = \{c\} \cup (L \setminus \{a, c\}) = L \setminus \{a\}$.
- $(a \boxplus c) \boxplus a = \bigcup_{p \in L \setminus \{a,c\}} (p \boxplus a) = \bigcup_{p \in L \setminus \{a,c\}} (L \setminus \{p, a\})$. The point a lies in none of these sets. A point $q \in L \setminus \{a\}$ is missing from $L \setminus \{p, a\}$ only when $p = q$, and there are at least 3 choices of p. So the union is $L \setminus \{a\}$.

Case 3. a, b, c are distinct and lie on a common line L. Here $a \boxplus b = L \setminus \{a, b\}$, which contains c. Then

$$S = (c \boxplus c) \cup \bigcup_{p \in L \setminus \{a,b,c\}} (L \setminus \{p, c\}) = \{0, c\} \cup (L \setminus \{c\}) = L \cup \{0\},$$

where we used that $L \setminus \{a, b, c\}$ has at least 2 points, by (P3). The result is symmetric in a, b, c.

Case 4. a, b, c are not collinear. We claim

$$S = G \setminus \big(\mathrm{L}(a, b) \cup \mathrm{L}(b, c) \cup \mathrm{L}(a, c)\big),$$

which is symmetric in a, b, c. By definition,

$$S = \bigcup_{p \in \mathrm{L}(a,b) \setminus \{a,b\}} \big(\mathrm{L}(p, c) \setminus \{p, c\}\big);$$

here $p \neq c$ because $c \notin \mathrm{L}(a, b)$. Also $\mathrm{L}(p, c) \neq \mathrm{L}(a, b)$, so $\mathrm{L}(p, c) \cap \mathrm{L}(a, b) = \{p\}$.

- $0 \notin S$, and $c \notin S$.
- *Points of* $\mathrm{L}(a, b)$ *are not in* S (including a and b): such a point lies in $\mathrm{L}(p, c)$ only if it equals p, and p is excluded.
- *Points* $q \in \mathrm{L}(a, c) \setminus \{a, c\}$ *are not in* S. If $q \in \mathrm{L}(p, c) \setminus \{p, c\}$, then $\mathrm{L}(p, c) = \mathrm{L}(q, c) = \mathrm{L}(a, c)$. So $p \in \mathrm{L}(a, c) \cap \mathrm{L}(a, b) = \{a\}$, since these lines are distinct; this contradicts $p \neq a$.
- *Points* $q \in \mathrm{L}(b, c) \setminus \{b, c\}$ *are not in* S, by the same argument with b in place of a.
- *Every other point* q *of* G *is in* S. Such a q lies on none of the three lines. The lines $\mathrm{L}(q, c)$ and $\mathrm{L}(a, b)$ are distinct because $c \notin \mathrm{L}(a, b)$, so by (P2) they meet in a point p. Then $p \neq a$, since otherwise $q \in \mathrm{L}(a, c)$; likewise $p \neq b$. Also $p \neq q$, since $q \notin \mathrm{L}(a, b)$, and $p \neq c$. So $q \in \mathrm{L}(p, c) \setminus \{p, c\}$ with $p \in \mathrm{L}(a, b) \setminus \{a, b\}$, hence $q \in S$.

These cases cover every multiset of three nonzero elements, so $\boxplus$ is associative. □

7 The six-term sum in such a hyperfield

Lemma 4. *Let* $\mathbb{H}$ *be as in Section 6, and let* $x, y \in G$ *satisfy:*
(i) *the points* $1, x, y$ *are distinct and collinear;*
(ii) *the points* x^3, y^3, xy *are distinct and lie on a line* $\mathcal{M}$ *with* $1 \notin \mathcal{M}$.
Then with $z = 1$ *we have* $0 \in x \boxplus y \boxplus z$, *but* $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$.

Proof. The hypothesis holds. Since $1, x, y$ are distinct and collinear, $1 \in \mathrm{L}(x,y) \setminus \{x,y\} = x \boxplus y$. Hence $(x \boxplus y) \boxplus 1 \supseteq 1 \boxplus 1 \ni 0$.

The conclusion fails. Every element is its own negative (Proposition 3), so $-xyz = xy$; also $z^3 = 1$. The six-term sum is therefore $T \boxplus 1$, where

$$T := x^3 \boxplus y^3 \boxplus xy \boxplus xy \boxplus xy.$$

First, $xy \boxplus xy \boxplus xy = \{0, xy\} \boxplus xy = \{xy\} \cup \{0, xy\} = \{0, xy\}$. So

$$T = (x^3 \boxplus y^3) \boxplus \{0, xy\} = (x^3 \boxplus y^3) \cup \big((x^3 \boxplus y^3) \boxplus xy\big).$$

The first part is $\mathcal{M} \setminus \{x^3, y^3\}$. The second part is $\mathcal{M} \cup \{0\}$, by Case 3 of the proof of Proposition 3, since x^3, y^3, xy are distinct and lie on $\mathcal{M}$. Hence $T = \mathcal{M} \cup \{0\}$.

Finally, $T \boxplus 1 = \bigcup_{u \in T} u \boxplus 1$. By uniqueness of negatives, $0 \in u \boxplus 1$ if and only if $u = -1 = 1$. Since $1 \notin \mathcal{M}$ and $1 \neq 0$, we have $1 \notin T$, so $0 \notin T \boxplus 1$. □

8 A plane containing the required configuration

From now on $G = \mathbb{Z}$, written additively: the integer n stands for the group element t^n, so the identity element 1 of G corresponds to $0 \in \mathbb{Z}$, and multiplication in G corresponds to addition in $\mathbb{Z}$.

Definition 5. A set $D \subset \mathbb{Z}$ is a *perfect difference set* if every nonzero integer has exactly one representation $d - d'$ with $d, d' \in D$. A set $D \subset \mathbb{Z}$ is *Sidon* if every nonzero integer has at most one such representation.

Lemma 6. *If $D \subset \mathbb{Z}$ is an infinite perfect difference set, then the lines $\{g + D : g \in \mathbb{Z}\}$ on the point set $\mathbb{Z}$ satisfy* (P1)–(P4).

Proof. (P1): Distinct points p, q lie on $g + D$ if and only if $p - g = d_1$ and $q - g = d_2$ for some $d_1, d_2 \in D$. Then $p - q = d_1 - d_2$. This representation of the nonzero integer $p - q$ exists and is unique, so $g = p - d_1$ exists and is unique.

(P2): Distinct lines are $g + D$ and $h + D$ with $g \neq h$. Let $h - g = d_1 - d_2$ be the representation. Then both lines contain the point $g + d_1 = h + d_2$.

(P3): Every line is infinite.

(P4): Translating $g + D$ by w gives $(g + w) + D$, which is again a line. □

Lemma 7. *There is an infinite perfect difference set $D \subset \mathbb{Z}$ with*

$$D \supseteq D_0 := \{0, 1, 4, 103, 105, 112\} \qquad \textit{and} \qquad 100 \notin D.$$

> **Reviewer 1**
>
> This lemma is essentially [3, Theorem 3.1]. Again, this paper was mentioned in the introduction, but it should be explicitly mentioned here. It should also be mentioned explicitly that the argument here do some minor extra work to ensure $100 \notin D$. The argument could have instead used $4 - 1 = 103 - 100$ to rule out the possibility that $100 \in D$.

Proof. D_0 is Sidon. Its 15 positive differences are

$$1 - 0 = 1, \quad 4 - 0 = 4, \quad 4 - 1 = 3, \quad 103 - 0 = 103, \quad 103 - 1 = 102, \quad 103 - 4 = 99,$$
$$105 - 0 = 105, \quad 105 - 1 = 104, \quad 105 - 4 = 101, \quad 105 - 103 = 2,$$
$$112 - 0 = 112, \quad 112 - 1 = 111, \quad 112 - 4 = 108, \quad 112 - 103 = 9, \quad 112 - 105 = 7,$$

that is, $1, 2, 3, 4, 7, 9, 99, 101, 102, 103, 104, 105, 108, 111, 112$. These are pairwise distinct.

Inductive step. Let D_k be a finite Sidon set of nonnegative integers, and let

- m be the least positive integer that is not a difference of two elements of D_k;
- $R = \max D_k$;
- N be an integer with $N > 3R + m$.

Put $D_{k+1} = D_k \cup \{N, N+m\}$. These are new elements, since $N > R$. The new positive differences are $m = (N+m) - N$, the numbers $N - d$, and the numbers $N + m - d$, for $d \in D_k$. We check that all positive differences of D_{k+1} (counted with their representations) are distinct:

- Old positive differences are at most R. Every $N - d$ and every $N + m - d$ is at least $N - R > 2R + m$. So these exceed both the old differences and m.
- The values $N - d$ ($d \in D_k$) are pairwise distinct, and so are the values $N + m - d$.
- If $N - d = N + m - d'$, then $d' - d = m$, which would make m a difference of D_k; this is impossible.
- m itself is not an old difference, by its choice.

So D_{k+1} is a Sidon set of nonnegative integers, and m is now a difference.

Covering. Every difference of D_k remains a difference of D_{k+1}, and m becomes one. So the least positive non-difference strictly increases at each step. Hence every positive integer is eventually a difference, and by symmetry so is every negative integer.

The limit set. Let $D = \bigcup_k D_k$. It is infinite. Any two representations of the same nonzero integer as a difference of elements of D already lie in a common D_k, so they coincide. By covering, every nonzero integer has at least one representation. So D is a perfect difference set.

$100 \notin D$. We have $100 \notin D_0$, and every element added at a later step is at least $N > 3R \geq 3 \cdot 112 > 100$, as $R \geq \max D_0 = 112$ at every step. □

9 The counterexample

Proof of Theorem 1. Let D be as in Lemma 7. By Lemma 6, the lines $g + D$ on $\mathbb{Z}$ satisfy (P1)–(P4), so Proposition 3 gives a hyperfield $\mathbb{H} = \langle t \rangle \cup \{0\}$. Take $x = t$, $y = t^4$, $z = 1$; in additive notation $x \leftrightarrow 1$, $y \leftrightarrow 4$ and $1_{\mathbb{H}} \leftrightarrow 0$. We check the hypotheses of Lemma 4.

- The points $1_{\mathbb{H}}, x, y$ are $0, 1, 4$. They are distinct and all lie on the line D.
- The points x^3, y^3, xy are $3, 12, 5$. They are distinct and lie on the line $\mathcal{M} = -100 + D$, because $103, 112, 105 \in D$.
- $1_{\mathbb{H}}$ is not on $\mathcal{M}$: the point 0 lies on $\mathcal{M}$ if and only if $100 \in D$, which is false.

By Lemma 4, $0 \in x \boxplus y \boxplus z$ but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$. This proves Theorem 1, so the answer to Question 1 is no. □

Remark 8. The following observation is not used in the proof; it explains why a counterexample cannot be of the most familiar type. Let F be a commutative field and $K^\times$ a subgroup of $F^\times$, and consider Krasner's quotient hyperfield $F/K^\times$ [5], with classes $[X] = XK^\times$ and $[X] \boxplus [Y] = \{[X' + Y'] : X' \in [X], Y' \in [Y]\}$. Suppose $0 \in x \boxplus y \boxplus z$ there. Then some $w \in x \boxplus y$ satisfies $0 \in w \boxplus z$, so $w = -z$; hence there are representatives $X \in x$, $Y \in y$ with $X + Y \in -z$, and $Z := -(X + Y)$ is a representative of z with $X + Y + Z = 0$. The identity

$$X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$$

shows that $X^3 + Y^3 + Z^3 + (-XYZ) + (-XYZ) + (-XYZ) = 0$, and $X^3, Y^3, Z^3, -XYZ$ are representatives of $x^3, y^3, z^3, -xyz$; so 0 lies in the six-term sum. Heuristically, this is why the counterexample above uses a projective plane that is not of field origin. We do not prove here that the plane of Lemma 7 is non-Desarguesian, and nothing depends on it.

References

[1] M. Baker and N. Bowler, Matroids over partial hyperstructures, *Adv. Math.* **343** (2019), 821–863.

[2] A. Connes and C. Consani, The hyperring of adèle classes, *J. Number Theory* **131** (2011), 159–194.

[3] M. Hall, Jr., Cyclic projective planes, *Duke Math. J.* **14** (1947), 1079–1090.

[4] M. Krasner, Approximation des corps valués complets de caractéristique $p \neq 0$ par ceux de caractéristique 0, in: *Colloque d'algèbre supérieure (Bruxelles, 1956)*, 1957.

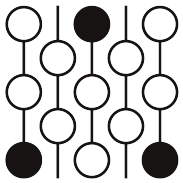
[5] M. Krasner, A class of hyperrings and hyperfields, *Internat. J. Math. Math. Sci.* **6** (1983), 307–311.

[6] O. Viro, Hyperfields for tropical geometry I. Hyperfields and dequantization, preprint, arXiv:1006.3034 (2010).

Review of Problem 5 – Siege

First Proof Third Batch
Reviewer 2

October 8–9, 2026



1 Review

This problem turns out to admit a solution using existing work in the literature. Essentially, a solution may be found by combining a paper of Connes & Consani and a paper of Hall; the former of which is directly about hyperfields. Once these two papers have been found, no great effort is required to solve the problem, more on the level of a challenging homework problem (perhaps 2 pages once definitions have been given) than a research question. The review should be read in that light.

The mathematics here are correct. The exposition is pretty good. I didn't struggle to read any of the arguments, although there are some odd notation choices. The two key citations of Connes & Consani, and Hall are included in the overview but totally absent from the text itself, including in places where the arguments of those authors are produced almost verbatim. This makes it quite obscure what is novel in the paper, affecting the exposition and attribution score.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

3 The problem

Question 1. Let $\mathbb{H}$ be a hyperfield with multivalued addition operator $\boxplus$. Let $x, y, z \in \mathbb{H}$ be elements. Suppose that $0 \in x \boxplus y \boxplus z$. Does it then follow that

$$0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus -xyz \boxplus -xyz \boxplus -xyz\,?$$

4 The result

The answer is *no*.

Theorem 1. *There is a hyperfield $\mathbb{H}$ with elements x, y, z such that*

$$0 \in x \boxplus y \boxplus z \qquad \textit{but} \qquad 0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

One may take $\mathbb{H}$ infinite, with multiplicative group $\mathbb{H}^\times = \langle t \rangle \cong \mathbb{Z}$ infinite cyclic, and $x = t$, $y = t^4$, $z = 1$.

Conventions. "Hyperfield" is meant in the standard sense of Krasner [4, 5]; see also [6, 1]. Thus a hyperfield is a set $\mathbb{H}$ with a distinguished element 0 and the following structure:

- a commutative, associative hyperoperation $\boxplus$ with nonempty values, for which 0 is neutral;
- unique negatives: for each a there is exactly one b, denoted $-a$, with $0 \in a \boxplus b$;
- reversibility: $c \in a \boxplus b \Rightarrow a \in c \boxplus (-b)$;
- a multiplication making $\mathbb{H} \setminus \{0\}$ an abelian group, with $0 \cdot h = h \cdot 0 = 0$ and $w(a \boxplus b) = wa \boxplus wb$.

Sums of sets are taken elementwise, $A \boxplus B = \bigcup_{a \in A,\, b \in B} a \boxplus b$, and $a \boxplus B := \{a\} \boxplus B$. By associativity, iterated sums do not depend on the bracketing (nor, by commutativity, on the order of the summands).

> **Reviewer 2**
>
> The cited definitions don't totally agree with one another, better to be explicit about which conventions are used.

5 Overview of the proof

When $\mathbb{H}$ is a quotient $F/K^\times$ of a field, the answer is yes, because of the identity $X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$ (Remark 8). A counterexample must therefore come from a hyperfield that is not of this kind, and we use hyperfields built from projective planes.

Given an abelian group G acting regularly on the points of a projective plane by collineations, with at least five points on each line, set $\mathbb{H} = G \cup \{0\}$, $a \boxplus a = \{0, a\}$ and $a \boxplus b = \mathrm{L}(a, b) \setminus \{a, b\}$ for $a \neq b$, where $\mathrm{L}(a, b)$ is the line through a and b. This is a hyperfield in which every element is its own negative (Proposition 3); a construction of this kind appears in the work of Connes and Consani [2]. In such an $\mathbb{H}$, with $z = 1$, the hypothesis $0 \in x \boxplus y \boxplus 1$ holds as soon as $1, x, y$ are distinct and collinear, while the six-term sum equals $(\mathcal{M} \cup \{0\}) \boxplus 1$, where $\mathcal{M}$ is the line through x^3, y^3, xy (if these are distinct). This sum avoids 0 whenever $1 \notin \mathcal{M}$ (Lemma 4). So we need a plane in which $1, x, y$ are collinear but 1 is not on the line through x^3, y^3, xy; by the remark above, no quotient of a field can serve here.

Such a plane is obtained on the point set $\mathbb{Z}$ (written additively, so that n stands for t^n) from an infinite perfect difference set $D \subset \mathbb{Z}$, whose translates $g + D$ serve as lines (Lemma 6). A greedy construction (Lemma 7), in the spirit of M. Hall's infinite cyclic planes [3], produces such a D containing the seed $\{0, 1, 4, 103, 105, 112\}$ and not containing 100. Then $0, 1, 4$ lie on the line D, the points $3, 12, 5$ (corresponding to x^3, y^3, xy) lie on the line $-100 + D$, and 0 does not.

All arguments below are given in full; the only external input is the definition of a hyperfield.

6 Hyperfields from projective planes with a regular abelian group

Let G be an abelian group, written multiplicatively. Consider an incidence structure with point set G and a family of subsets of G, called *lines*, satisfying:
(P1) any two distinct points lie on exactly one line;
(P2) any two distinct lines meet;
(P3) every line has at least 5 points;
(P4) for every $w \in G$, multiplication by w maps lines to lines.
By (P1) two distinct lines have at most one common point, and by (P2) they have at least one; so two distinct lines meet in exactly one point. For distinct points a, b we write $\mathrm{L}(a, b)$ for the line through a and b.

> **Reviewer 2**
>
> The definitions and arguments in this section are basically identical to those in Connes and Consani, which is mentioned in the overview, but really deserves in line citations since they are so similar.

Definition 2. Put $\mathbb{H} = G \cup \{0\}$. Let 0 be neutral: $0 \boxplus a = a \boxplus 0 = \{a\}$ for all $a \in \mathbb{H}$ (so $0 \boxplus 0 = \{0\}$). For $a, b \in G$ set

$$a \boxplus a = \{0, a\}, \qquad a \boxplus b = \mathrm{L}(a, b) \setminus \{a, b\} \quad (a \neq b).$$

Multiplication is the group law on G, extended by $0 \cdot h = h \cdot 0 = 0$.

Proposition 3. *$\mathbb{H}$ is a hyperfield, and $-a = a$ for every $a \in \mathbb{H}$.*

Proof. Commutativity and nonempty values. Commutativity is clear. For $a \neq b$ in G, the set $a \boxplus b$ has at least 3 points by (P3), so every value is nonempty.

Negatives. Let $a \neq 0$. For $b \in G$ with $b \neq a$ we have $a \boxplus b \subseteq G$, so $0 \notin a \boxplus b$. Also $0 \boxplus a = \{a\}$ does not contain 0, while $0 \in a \boxplus a$. Hence a is the unique negative of a. The element 0 is its own unique negative.

Reversibility. Since $-b = b$, we must show $c \in a \boxplus b \Rightarrow a \in c \boxplus b$.

- If $a = 0$, then $c = b$ and $0 \in b \boxplus b$; this includes the case $b = 0$.
- If $b = 0$, then $c = a$ and $a \in a \boxplus 0$.
- If $a = b \neq 0$, then $c \in \{0, a\}$. If $c = 0$, then $a \in 0 \boxplus a$. If $c = a$, then $a \in a \boxplus a$.
- If $a \neq b$, both nonzero, then $c \in \mathrm{L}(a, b) \setminus \{a, b\}$. Since $c \neq b$, $\mathrm{L}(c, b) = \mathrm{L}(a, b)$, so $c \boxplus b = \mathrm{L}(a, b) \setminus \{c, b\}$. This set contains a, because $a \neq c$ and $a \neq b$.

Distributivity. The case $w = 0$ is trivial, so let $w \in G$. Multiplication by w is a bijection of G that maps lines to lines by (P4). Hence $w\,\mathrm{L}(a, b)$ is a line through wa and wb, i.e. $w\,\mathrm{L}(a, b) = \mathrm{L}(wa, wb)$, and so $w(\mathrm{L}(a, b) \setminus \{a, b\}) = \mathrm{L}(wa, wb) \setminus \{wa, wb\}$. Also $w\{0, a\} = \{0, wa\} = wa \boxplus wa$, and the cases involving 0 are trivial.

Associativity. If one of a, b, c is 0, both $(a \boxplus b) \boxplus c$ and $a \boxplus (b \boxplus c)$ equal the sum of the other two, because 0 is neutral.

> **Reviewer 2**
>
> I think the following case organization is maybe the clearest of the solutions I read

Now let a, b, c be nonzero and put $S(a, b, c) := (a \boxplus b) \boxplus c$. We show that S depends only on the multiset $\{a, b, c\}$. Then, using commutativity,

$$a \boxplus (b \boxplus c) = (b \boxplus c) \boxplus a = S(b, c, a) = S(a, b, c) = (a \boxplus b) \boxplus c.$$

Case 1. $a = b = c$. Then $S = (0 \boxplus a) \cup (a \boxplus a) = \{0, a\}$.

Case 2. The multiset is $\{a, a, c\}$ with $a \neq c$; let $L = \mathrm{L}(a, c)$. Up to commutativity there are two orderings.

- $(a \boxplus a) \boxplus c = (0 \boxplus c) \cup (a \boxplus c) = \{c\} \cup (L \setminus \{a, c\}) = L \setminus \{a\}$.
- $(a \boxplus c) \boxplus a = \bigcup_{p \in L \setminus \{a,c\}} (p \boxplus a) = \bigcup_{p \in L \setminus \{a,c\}} (L \setminus \{p, a\})$. The point a lies in none of these sets. A point $q \in L \setminus \{a\}$ is missing from $L \setminus \{p, a\}$ only when $p = q$, and there are at least 3 choices of p. So the union is $L \setminus \{a\}$.

Case 3. a, b, c are distinct and lie on a common line L. Here $a \boxplus b = L \setminus \{a, b\}$, which contains c. Then

$$S = (c \boxplus c) \cup \bigcup_{p \in L \setminus \{a,b,c\}} (L \setminus \{p, c\}) = \{0, c\} \cup (L \setminus \{c\}) = L \cup \{0\},$$

where we used that $L \setminus \{a, b, c\}$ has at least 2 points, by (P3). The result is symmetric in a, b, c.

Case 4. a, b, c are not collinear. We claim

$$S = G \setminus \big(\mathrm{L}(a, b) \cup \mathrm{L}(b, c) \cup \mathrm{L}(a, c)\big),$$

which is symmetric in a, b, c. By definition,

$$S = \bigcup_{p \in \mathrm{L}(a,b) \setminus \{a,b\}} \big(\mathrm{L}(p, c) \setminus \{p, c\}\big);$$

here $p \neq c$ because $c \notin \mathrm{L}(a, b)$. Also $\mathrm{L}(p, c) \neq \mathrm{L}(a, b)$, so $\mathrm{L}(p, c) \cap \mathrm{L}(a, b) = \{p\}$.

- $0 \notin S$, and $c \notin S$.
- *Points of* $\mathrm{L}(a, b)$ *are not in* S (including a and b): such a point lies in $\mathrm{L}(p, c)$ only if it equals p, and p is excluded.
- *Points* $q \in \mathrm{L}(a, c) \setminus \{a, c\}$ *are not in* S. If $q \in \mathrm{L}(p, c) \setminus \{p, c\}$, then $\mathrm{L}(p, c) = \mathrm{L}(q, c) = \mathrm{L}(a, c)$. So $p \in \mathrm{L}(a, c) \cap \mathrm{L}(a, b) = \{a\}$, since these lines are distinct; this contradicts $p \neq a$.
- *Points* $q \in \mathrm{L}(b, c) \setminus \{b, c\}$ *are not in* S, by the same argument with b in place of a.
- *Every other point* q *of* G *is in* S. Such a q lies on none of the three lines. The lines $\mathrm{L}(q, c)$ and $\mathrm{L}(a, b)$ are distinct because $c \notin \mathrm{L}(a, b)$, so by (P2) they meet in a point p. Then $p \neq a$, since otherwise $q \in \mathrm{L}(a, c)$; likewise $p \neq b$. Also $p \neq q$, since $q \notin \mathrm{L}(a, b)$, and $p \neq c$. So $q \in \mathrm{L}(p, c) \setminus \{p, c\}$ with $p \in \mathrm{L}(a, b) \setminus \{a, b\}$, hence $q \in S$.

These cases cover every multiset of three nonzero elements, so $\boxplus$ is associative. □

7 The six-term sum in such a hyperfield

Lemma 4. *Let $\mathbb{H}$ be as in Section 6, and let $x, y \in G$ satisfy:*
(i) *the points $1, x, y$ are distinct and collinear;*
(ii) *the points x^3, y^3, xy are distinct and lie on a line $\mathcal{M}$ with $1 \notin \mathcal{M}$.*
Then with $z = 1$ we have $0 \in x \boxplus y \boxplus z$, but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$.

Proof. The hypothesis holds. Since $1, x, y$ are distinct and collinear, $1 \in \mathrm{L}(x, y) \setminus \{x, y\} = x \boxplus y$. Hence $(x \boxplus y) \boxplus 1 \supseteq 1 \boxplus 1 \ni 0$.

The conclusion fails. Every element is its own negative (Proposition 3), so $-xyz = xy$; also $z^3 = 1$. The six-term sum is therefore $T \boxplus 1$, where

$$T := x^3 \boxplus y^3 \boxplus xy \boxplus xy \boxplus xy.$$

First, $xy \boxplus xy \boxplus xy = \{0, xy\} \boxplus xy = \{xy\} \cup \{0, xy\} = \{0, xy\}$. So

$$T = (x^3 \boxplus y^3) \boxplus \{0, xy\} = (x^3 \boxplus y^3) \cup \big((x^3 \boxplus y^3) \boxplus xy\big).$$

The first part is $\mathcal{M} \setminus \{x^3, y^3\}$. The second part is $\mathcal{M} \cup \{0\}$, by Case 3 of the proof of Proposition 3, since x^3, y^3, xy are distinct and lie on $\mathcal{M}$. Hence $T = \mathcal{M} \cup \{0\}$.

Finally, $T \boxplus 1 = \bigcup_{u \in T} u \boxplus 1$. By uniqueness of negatives, $0 \in u \boxplus 1$ if and only if $u = -1 = 1$. Since $1 \notin \mathcal{M}$ and $1 \neq 0$, we have $1 \notin T$, so $0 \notin T \boxplus 1$. □

8 A plane containing the required configuration

From now on $G = \mathbb{Z}$, written additively: the integer n stands for the group element t^n, so the identity element 1 of G corresponds to $0 \in \mathbb{Z}$, and multiplication in G corresponds to addition in $\mathbb{Z}$.

Definition 5. A set $D \subset \mathbb{Z}$ is a *perfect difference set* if every nonzero integer has exactly one representation $d - d'$ with $d, d' \in D$. A set $D \subset \mathbb{Z}$ is *Sidon* if every nonzero integer has at most one such representation.

Lemma 6. *If $D \subset \mathbb{Z}$ is an infinite perfect difference set, then the lines $\{g + D : g \in \mathbb{Z}\}$ on the point set $\mathbb{Z}$ satisfy* (P1)–(P4).

Proof. (P1): Distinct points p, q lie on $g + D$ if and only if $p - g = d_1$ and $q - g = d_2$ for some $d_1, d_2 \in D$. Then $p - q = d_1 - d_2$. This representation of the nonzero integer $p - q$ exists and is unique, so $g = p - d_1$ exists and is unique.

(P2): Distinct lines are $g + D$ and $h + D$ with $g \neq h$. Let $h - g = d_1 - d_2$ be the representation. Then both lines contain the point $g + d_1 = h + d_2$.

(P3): Every line is infinite.

(P4): Translating $g + D$ by w gives $(g + w) + D$, which is again a line. □

Lemma 7. *There is an infinite perfect difference set $D \subset \mathbb{Z}$ with*

$$D \supseteq D_0 := \{0, 1, 4, 103, 105, 112\} \qquad \textit{and} \qquad 100 \notin D.$$

Proof. D_0 is Sidon. Its 15 positive differences are

$$1 - 0 = 1, \quad 4 - 0 = 4, \quad 4 - 1 = 3, \quad 103 - 0 = 103, \quad 103 - 1 = 102, \quad 103 - 4 = 99,$$
$$105 - 0 = 105, \quad 105 - 1 = 104, \quad 105 - 4 = 101, \quad 105 - 103 = 2,$$
$$112 - 0 = 112, \quad 112 - 1 = 111, \quad 112 - 4 = 108, \quad 112 - 103 = 9, \quad 112 - 105 = 7,$$

that is, $1, 2, 3, 4, 7, 9, 99, 101, 102, 103, 104, 105, 108, 111, 112$. These are pairwise distinct.

Inductive step. Let D_k be a finite Sidon set of nonnegative integers, and let

- m be the least positive integer that is not a difference of two elements of D_k;
- $R = \max D_k$;
- N be an integer with $N > 3R + m$.

Put $D_{k+1} = D_k \cup \{N, N+m\}$. These are new elements, since $N > R$. The new positive differences are $m = (N+m) - N$, the numbers $N - d$, and the numbers $N + m - d$, for $d \in D_k$. We check that all positive differences of D_{k+1} (counted with their representations) are distinct:

- Old positive differences are at most R. Every $N - d$ and every $N + m - d$ is at least $N - R > 2R + m$. So these exceed both the old differences and m.
- The values $N - d$ ($d \in D_k$) are pairwise distinct, and so are the values $N + m - d$.
- If $N - d = N + m - d'$, then $d' - d = m$, which would make m a difference of D_k; this is impossible.
- m itself is not an old difference, by its choice.

So D_{k+1} is a Sidon set of nonnegative integers, and m is now a difference.

Covering. Every difference of D_k remains a difference of D_{k+1}, and m becomes one. So the least positive non-difference strictly increases at each step. Hence every positive integer is eventually a difference, and by symmetry so is every negative integer.

The limit set. Let $D = \bigcup_k D_k$. It is infinite. Any two representations of the same nonzero integer as a difference of elements of D already lie in a common D_k, so they coincide. By covering, every nonzero integer has at least one representation. So D is a perfect difference set.

$100 \notin D$. We have $100 \notin D_0$, and every element added at a later step is at least $N > 3R \geq 3 \cdot 112 > 100$, as $R \geq \max D_0 = 112$ at every step. □

> **Reviewer 2**
>
> This is a known method due to Hall 1947 and should be cited. Indeed the proof is extremely similar to Hall's and it would be useful to highlight where it differs.

9 The counterexample

Proof of Theorem 1. Let D be as in Lemma 7. By Lemma 6, the lines $g + D$ on $\mathbb{Z}$ satisfy (P1)–(P4), so Proposition 3 gives a hyperfield $\mathbb{H} = \langle t \rangle \cup \{0\}$. Take $x = t$, $y = t^4$, $z = 1$; in additive notation $x \leftrightarrow 1$, $y \leftrightarrow 4$ and $1_{\mathbb{H}} \leftrightarrow 0$. We check the hypotheses of Lemma 4.

- The points $1_{\mathbb{H}}, x, y$ are $0, 1, 4$. They are distinct and all lie on the line D.
- The points x^3, y^3, xy are $3, 12, 5$. They are distinct and lie on the line $\mathcal{M} = -100 + D$, because $103, 112, 105 \in D$.
- $1_{\mathbb{H}}$ is not on $\mathcal{M}$: the point 0 lies on $\mathcal{M}$ if and only if $100 \in D$, which is false.

By Lemma 4, $0 \in x \boxplus y \boxplus z$ but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$. This proves Theorem 1, so the answer to Question 1 is no. □

Remark 8. The following observation is not used in the proof; it explains why a counterexample cannot be of the most familiar type. Let F be a commutative field and $K^\times$ a subgroup of $F^\times$, and consider Krasner's quotient hyperfield $F/K^\times$ [5], with classes $[X] = XK^\times$ and $[X] \boxplus [Y] = \{[X' + Y'] : X' \in [X], Y' \in [Y]\}$. Suppose $0 \in x \boxplus y \boxplus z$ there. Then some $w \in x \boxplus y$ satisfies $0 \in w \boxplus z$, so $w = -z$; hence there are representatives $X \in x$, $Y \in y$ with $X + Y \in -z$, and $Z := -(X + Y)$ is a representative of z with $X + Y + Z = 0$. The identity

$$X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$$

shows that $X^3 + Y^3 + Z^3 + (-XYZ) + (-XYZ) + (-XYZ) = 0$, and $X^3, Y^3, Z^3, -XYZ$ are representatives of $x^3, y^3, z^3, -xyz$; so 0 lies in the six-term sum. Heuristically, this is why the counterexample above uses a projective plane that is not of field origin. We do not prove here that the plane of Lemma 7 is non-Desarguesian, and nothing depends on it.

References

[1] M. Baker and N. Bowler, Matroids over partial hyperstructures, *Adv. Math.* **343** (2019), 821–863.

[2] A. Connes and C. Consani, The hyperring of adèle classes, *J. Number Theory* **131** (2011), 159–194.

[3] M. Hall, Jr., Cyclic projective planes, *Duke Math. J.* **14** (1947), 1079–1090.

[4] M. Krasner, Approximation des corps valués complets de caractéristique $p \neq 0$ par ceux de caractéristique 0, in: *Colloque d'algèbre supérieure (Bruxelles, 1956)*, 1957.

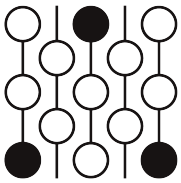
[5] M. Krasner, A class of hyperrings and hyperfields, *Internat. J. Math. Math. Sci.* **6** (1983), 307–311.

[6] O. Viro, Hyperfields for tropical geometry I. Hyperfields and dequantization, preprint, arXiv:1006.3034 (2010).

Review of Problem 5 – Siege

First Proof Third Batch
Reviewer 3

October 8–9, 2026



1 Review

In a ring or a field, the identity $x + y + z = 0$ implies the identity $x^3 + y^3 + z^3 - 3xyz = 0$. This submitted note provides a counterexample for its counterpart for hyperfields, in which the addition is possibly multi-valued.

The counterexample presented in this note follows the ideas of Hall, Prenowitz, and Connes–Consani. Concretely, Prenowitz and Connes–Consani connected hyperfields with point-line arrangements, and Hall gave constructions of an infinite point-line arrangement from $\mathbb{Z}$. I do not find any mistakes in the proof. However, the historical remarks are only included in the overview of the proof and not in the main body of the submission when specific constructions, lemmas, and theorems are given. The construction of a hyperfield from a projective plane (Proposition 3) is almost identical to the construction and proof in Connes–Consani, and the extension construction of the point-line arrangements (Lemma 7) is almost identical to the construction and proof in Hall.

The organization of the paper is somewhat confusing. Section 6 and Section 7 should both give the readers some hints for the existence of the incidence structure and the existence of elements x and y. An alternate fix is to first fix $G = \mathbb{Z}$ as in Section 8 and give all the relevant results on perfect difference sets, and then construct projective planes and hyperfields from this particular G.

For publication in a journal, I recommend a major revision. In particular, the author should remove the proofs that are already written in the literature and only cite the corresponding papers.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort

- -1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

3 The problem

Question 1. Let $\mathbb{H}$ be a hyperfield with multivalued addition operator $\boxplus$. Let $x, y, z \in \mathbb{H}$ be elements. Suppose that $0 \in x \boxplus y \boxplus z$. Does it then follow that

$$0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus -xyz \boxplus -xyz \boxplus -xyz\,?$$

4 The result

The answer is *no*.

Theorem 1. *There is a hyperfield $\mathbb{H}$ with elements x, y, z such that*

$$0 \in x \boxplus y \boxplus z \qquad \textit{but} \qquad 0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz).$$

One may take $\mathbb{H}$ infinite, with multiplicative group $\mathbb{H}^\times = \langle t \rangle \cong \mathbb{Z}$ infinite cyclic, and $x = t$, $y = t^4$, $z = 1$.

Conventions. "Hyperfield" is meant in the standard sense of Krasner [4, 5]; see also [6, 1]. Thus a hyperfield is a set $\mathbb{H}$ with a distinguished element 0 and the following structure:

> **Reviewer 3**
> For exposition, the definition of hyperfields should be clearly stated instead of in "Conventions".

- a commutative, associative hyperoperation $\boxplus$ with nonempty values, for which 0 is neutral;

> **Reviewer 3**
> Need to specify the source and target of hyperoperation. Need to explain what "0 is neutral" means. Need to define commutativity and associativity for hyperoperations.

- unique negatives: for each a there is exactly one b, denoted $-a$, with $0 \in a \boxplus b$;

> **Reviewer 3**
> Where do a and b live?

- reversibility: $c \in a \boxplus b \Rightarrow a \in c \boxplus (-b)$;

> **Reviewer 3**
> Where do a, b, c live? The more common definition in the literature for the reversibility is an if and only if statement. The notation "$\Rightarrow$" should be replaced by the corresponding words in English.

- a multiplication making $\mathbb{H} \setminus \{0\}$ an abelian group, with $0 \cdot h = h \cdot 0 = 0$ and $w(a \boxplus b) = wa \boxplus wb$.

Sums of sets are taken elementwise, $A \boxplus B = \bigcup_{a \in A,\, b \in B} a \boxplus b$, and $a \boxplus B := \{a\} \boxplus B$. By associativity, iterated sums do not depend on the bracketing (nor, by commutativity, on the order of the summands).

> **Reviewer 3**
> Where do h, w, a, b live?

5 Overview of the proof

When $\mathbb{H}$ is a quotient $F/K^\times$ of a field, the answer is yes, because of the identity $X^3+Y^3+Z^3-3XYZ=(X+Y+Z)(X^2+Y^2+Z^2-XY-YZ-ZX)$ (Remark 8). A counterexample must therefore come from a hyperfield that is not of this kind, and we use hyperfields built from projective planes.

Given an abelian group G acting regularly on the points of a projective plane by collineations, with at least five points on each line, set $\mathbb{H}=G\cup\{0\}$, $a\boxplus a=\{0,a\}$ and $a\boxplus b=\mathrm{L}(a,b)\setminus\{a,b\}$ for $a\neq b$, where $\mathrm{L}(a,b)$ is the line through a and b. This is a hyperfield in which every element is its own negative (Proposition 3); a construction of this kind appears in the work of Connes and Consani [2]. In such an $\mathbb{H}$, with $z=1$, the hypothesis $0\in x\boxplus y\boxplus 1$ holds as soon as $1,x,y$ are distinct and collinear, while the six-term sum equals $(\mathcal{M}\cup\{0\})\boxplus 1$, where $\mathcal{M}$ is the line through x^3,y^3,xy (if these are distinct). This sum avoids 0 whenever $1\notin\mathcal{M}$ (Lemma 4). So we need a plane in which $1,x,y$ are collinear but 1 is not on the line through x^3,y^3,xy; by the remark above, no quotient of a field can serve here.

Such a plane is obtained on the point set $\mathbb{Z}$ (written additively, so that n stands for t^n) from an infinite perfect difference set $D\subset\mathbb{Z}$, whose translates $g+D$ serve as lines (Lemma 6). A greedy construction (Lemma 7), in the spirit of M. Hall's infinite cyclic planes [3], produces such a D containing the seed $\{0,1,4,103,105,112\}$ and not containing 100. Then $0,1,4$ lie on the line D, the points $3,12,5$ (corresponding to x^3,y^3,xy) lie on the line $-100+D$, and 0 does not.

All arguments below are given in full; the only external input is the definition of a hyperfield.

6 Hyperfields from projective planes with a regular abelian group

Let G be an abelian group, written multiplicatively. Consider an incidence structure with point set G and a family of subsets of G, called *lines*, satisfying:
(P1) any two distinct points lie on exactly one line;
(P2) any two distinct lines meet;
(P3) every line has at least 5 points;
(P4) for every $w\in G$, multiplication by w maps lines to lines.
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$$a\boxplus a=\{0,a\},\qquad a\boxplus b=\mathrm{L}(a,b)\setminus\{a,b\}\quad(a\neq b).$$

Multiplication is the group law on G, extended by $0\cdot h=h\cdot 0=0$.

Proposition 3. *$\mathbb{H}$ is a hyperfield, and $-a=a$ for every $a\in\mathbb{H}$.*

Proof. Commutativity and nonempty values. Commutativity is clear. For $a\neq b$ in G, the set $a\boxplus b$ has at least 3 points by (P3), so every value is nonempty.

Negatives. Let $a\neq 0$. For $b\in G$ with $b\neq a$ we have $a\boxplus b\subseteq G$, so $0\notin a\boxplus b$. Also $0\boxplus a=\{a\}$ does not contain 0, while $0\in a\boxplus a$. Hence a is the unique negative of a. The element 0 is its own unique negative.

Reversibility. Since $-b=b$, we must show $c\in a\boxplus b\Rightarrow a\in c\boxplus b$.

- If $a=0$, then $c=b$ and $0\in b\boxplus b$; this includes the case $b=0$.

- If $b = 0$, then $c = a$ and $a \in a \boxplus 0$.
- If $a = b \neq 0$, then $c \in \{0, a\}$. If $c = 0$, then $a \in 0 \boxplus a$. If $c = a$, then $a \in a \boxplus a$.
- If $a \neq b$, both nonzero, then $c \in \mathrm{L}(a, b) \setminus \{a, b\}$. Since $c \neq b$, $\mathrm{L}(c, b) = \mathrm{L}(a, b)$, so $c \boxplus b = \mathrm{L}(a, b) \setminus \{c, b\}$. This set contains a, because $a \neq c$ and $a \neq b$.

Distributivity. The case $w = 0$ is trivial, so let $w \in G$. Multiplication by w is a bijection of G that maps lines to lines by (P4). Hence $w\,\mathrm{L}(a, b)$ is a line through wa and wb, i.e. $w\,\mathrm{L}(a, b) = \mathrm{L}(wa, wb)$, and so $w(\mathrm{L}(a, b) \setminus \{a, b\}) = \mathrm{L}(wa, wb) \setminus \{wa, wb\}$. Also $w\{0, a\} = \{0, wa\} = wa \boxplus wa$, and the cases involving 0 are trivial.

Associativity. If one of a, b, c is 0, both $(a \boxplus b) \boxplus c$ and $a \boxplus (b \boxplus c)$ equal the sum of the other two, because 0 is neutral.

Now let a, b, c be nonzero and put $S(a, b, c) := (a \boxplus b) \boxplus c$. We show that S depends only on the multiset $\{a, b, c\}$. Then, using commutativity,

$$a \boxplus (b \boxplus c) = (b \boxplus c) \boxplus a = S(b, c, a) = S(a, b, c) = (a \boxplus b) \boxplus c.$$

Case 1. $a = b = c$. Then $S = (0 \boxplus a) \cup (a \boxplus a) = \{0, a\}$.

Case 2. The multiset is $\{a, a, c\}$ with $a \neq c$; let $L = \mathrm{L}(a, c)$. Up to commutativity there are two orderings.

- $(a \boxplus a) \boxplus c = (0 \boxplus c) \cup (a \boxplus c) = \{c\} \cup (L \setminus \{a, c\}) = L \setminus \{a\}$.
- $(a \boxplus c) \boxplus a = \bigcup_{p \in L \setminus \{a,c\}} (p \boxplus a) = \bigcup_{p \in L \setminus \{a,c\}} (L \setminus \{p, a\})$. The point a lies in none of these sets. A point $q \in L \setminus \{a\}$ is missing from $L \setminus \{p, a\}$ only when $p = q$, and there are at least 3 choices of p. So the union is $L \setminus \{a\}$.

Case 3. a, b, c are distinct and lie on a common line L. Here $a \boxplus b = L \setminus \{a, b\}$, which contains c. Then

$$S = (c \boxplus c) \cup \bigcup_{p \in L \setminus \{a,b,c\}} (L \setminus \{p, c\}) = \{0, c\} \cup (L \setminus \{c\}) = L \cup \{0\},$$

where we used that $L \setminus \{a, b, c\}$ has at least 2 points, by (P3). The result is symmetric in a, b, c.

Case 4. a, b, c are not collinear. We claim

$$S = G \setminus \big(\mathrm{L}(a, b) \cup \mathrm{L}(b, c) \cup \mathrm{L}(a, c)\big),$$

which is symmetric in a, b, c. By definition,

$$S = \bigcup_{p \in \mathrm{L}(a,b) \setminus \{a,b\}} \big(\mathrm{L}(p, c) \setminus \{p, c\}\big);$$

here $p \neq c$ because $c \notin \mathrm{L}(a, b)$. Also $\mathrm{L}(p, c) \neq \mathrm{L}(a, b)$, so $\mathrm{L}(p, c) \cap \mathrm{L}(a, b) = \{p\}$.

- $0 \notin S$, and $c \notin S$.
- *Points of* $\mathrm{L}(a, b)$ *are not in* S (including a and b): such a point lies in $\mathrm{L}(p, c)$ only if it equals p, and p is excluded.
- *Points* $q \in \mathrm{L}(a, c) \setminus \{a, c\}$ *are not in* S. If $q \in \mathrm{L}(p, c) \setminus \{p, c\}$, then $\mathrm{L}(p, c) = \mathrm{L}(q, c) = \mathrm{L}(a, c)$. So $p \in \mathrm{L}(a, c) \cap \mathrm{L}(a, b) = \{a\}$, since these lines are distinct; this contradicts $p \neq a$.
- *Points* $q \in \mathrm{L}(b, c) \setminus \{b, c\}$ *are not in* S, by the same argument with b in place of a.

- *Every other point q of G is in S.* Such a q lies on none of the three lines. The lines $\mathrm{L}(q,c)$ and $\mathrm{L}(a,b)$ are distinct because $c \notin \mathrm{L}(a,b)$, so by (P2) they meet in a point p. Then $p \neq a$, since otherwise $q \in \mathrm{L}(a,c)$; likewise $p \neq b$. Also $p \neq q$, since $q \notin \mathrm{L}(a,b)$, and $p \neq c$. So $q \in \mathrm{L}(p,c) \setminus \{p,c\}$ with $p \in \mathrm{L}(a,b) \setminus \{a,b\}$, hence $q \in S$.

These cases cover every multiset of three nonzero elements, so $\boxplus$ is associative. □

7 The six-term sum in such a hyperfield

Lemma 4. *Let $\mathbb{H}$ be as in Section 6, and let $x, y \in G$ satisfy:*
(i) *the points $1, x, y$ are distinct and collinear;*
(ii) *the points x^3, y^3, xy are distinct and lie on a line $\mathcal{M}$ with $1 \notin \mathcal{M}$.*
Then with $z = 1$ we have $0 \in x \boxplus y \boxplus z$, but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$.

Proof. The hypothesis holds. Since $1, x, y$ are distinct and collinear, $1 \in \mathrm{L}(x,y) \setminus \{x,y\} = x \boxplus y$. Hence $(x \boxplus y) \boxplus 1 \supseteq 1 \boxplus 1 \ni 0$.

The conclusion fails. Every element is its own negative (Proposition 3), so $-xyz = xy$; also $z^3 = 1$. The six-term sum is therefore $T \boxplus 1$, where

$$T := x^3 \boxplus y^3 \boxplus xy \boxplus xy \boxplus xy.$$

First, $xy \boxplus xy \boxplus xy = \{0, xy\} \boxplus xy = \{xy\} \cup \{0, xy\} = \{0, xy\}$. So

$$T = (x^3 \boxplus y^3) \boxplus \{0, xy\} = (x^3 \boxplus y^3) \cup \big((x^3 \boxplus y^3) \boxplus xy\big).$$

The first part is $\mathcal{M} \setminus \{x^3, y^3\}$. The second part is $\mathcal{M} \cup \{0\}$, by Case 3 of the proof of Proposition 3, since x^3, y^3, xy are distinct and lie on $\mathcal{M}$. Hence $T = \mathcal{M} \cup \{0\}$.

Finally, $T \boxplus 1 = \bigcup_{u \in T} u \boxplus 1$. By uniqueness of negatives, $0 \in u \boxplus 1$ if and only if $u = -1 = 1$. Since $1 \notin \mathcal{M}$ and $1 \neq 0$, we have $1 \notin T$, so $0 \notin T \boxplus 1$. □

8 A plane containing the required configuration

From now on $G = \mathbb{Z}$, written additively: the integer n stands for the group element t^n, so the identity element 1 of G corresponds to $0 \in \mathbb{Z}$, and multiplication in G corresponds to addition in $\mathbb{Z}$.

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> What is t?

Definition 5. A set $D \subset \mathbb{Z}$ is a *perfect difference set* if every nonzero integer has exactly one representation $d - d'$ with $d, d' \in D$. A set $D \subset \mathbb{Z}$ is *Sidon* if every nonzero integer has at most one such representation.

Lemma 6. *If $D \subset \mathbb{Z}$ is an infinite perfect difference set, then the lines $\{g + D : g \in \mathbb{Z}\}$ on the point set $\mathbb{Z}$ satisfy* (P1)–(P4).

Proof. (P1): Distinct points p, q lie on $g + D$ if and only if $p - g = d_1$ and $q - g = d_2$ for some $d_1, d_2 \in D$. Then $p - q = d_1 - d_2$. This representation of the nonzero integer $p - q$ exists and is unique, so $g = p - d_1$ exists and is unique.

(P2): Distinct lines are $g + D$ and $h + D$ with $g \neq h$. Let $h - g = d_1 - d_2$ be the representation. Then both lines contain the point $g + d_1 = h + d_2$.

(P3): Every line is infinite.

(P4): Translating $g + D$ by w gives $(g + w) + D$, which is again a line. □

Lemma 7. *There is an infinite perfect difference set $D \subset \mathbb{Z}$ with*

$$D \supseteq D_0 := \{0, 1, 4, 103, 105, 112\} \qquad \textit{and} \qquad 100 \notin D.$$

Proof. *D_0 is Sidon.* Its 15 positive differences are

$$1-0=1, \quad 4-0=4, \quad 4-1=3, \quad 103-0=103, \quad 103-1=102, \quad 103-4=99,$$
$$105-0=105, \quad 105-1=104, \quad 105-4=101, \quad 105-103=2,$$
$$112-0=112, \quad 112-1=111, \quad 112-4=108, \quad 112-103=9, \quad 112-105=7,$$

that is, $1, 2, 3, 4, 7, 9, 99, 101, 102, 103, 104, 105, 108, 111, 112$. These are pairwise distinct.

Inductive step. Let D_k be a finite Sidon set of nonnegative integers, and let

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> Should be clearer here what the induction is for. Lemma 7 as is is not a statement for k.

- m be the least positive integer that is not a difference of two elements of D_k;
- $R = \max D_k$;
- N be an integer with $N > 3R + m$.

Put $D_{k+1} = D_k \cup \{N, N+m\}$. These are new elements, since $N > R$. The new positive differences are $m = (N+m) - N$, the numbers $N - d$, and the numbers $N + m - d$, for $d \in D_k$. We check that all positive differences of D_{k+1} (counted with their representations) are distinct:

- Old positive differences are at most R. Every $N - d$ and every $N + m - d$ is at least $N - R > 2R + m$. So these exceed both the old differences and m.
- The values $N - d$ ($d \in D_k$) are pairwise distinct, and so are the values $N + m - d$.
- If $N - d = N + m - d'$, then $d' - d = m$, which would make m a difference of D_k; this is impossible.
- m itself is not an old difference, by its choice.

So D_{k+1} is a Sidon set of nonnegative integers, and m is now a difference.

Covering. Every difference of D_k remains a difference of D_{k+1}, and m becomes one. So the least positive non-difference strictly increases at each step. Hence every positive integer is eventually a difference, and by symmetry so is every negative integer.

The limit set. Let $D = \bigcup_k D_k$. It is infinite. Any two representations of the same nonzero integer as a difference of elements of D already lie in a common D_k, so they coincide. By covering, every nonzero integer has at least one representation. So D is a perfect difference set.

$100 \notin D$. We have $100 \notin D_0$, and every element added at a later step is at least $N > 3R \geq 3 \cdot 112 > 100$, as $R \geq \max D_0 = 112$ at every step. □

9 The counterexample

Proof of Theorem 1. Let D be as in Lemma 7. By Lemma 6, the lines $g + D$ on $\mathbb{Z}$ satisfy (P1)–(P4), so Proposition 3 gives a hyperfield $\mathbb{H} = \langle t \rangle \cup \{0\}$. Take $x = t$, $y = t^4$, $z = 1$; in additive notation $x \leftrightarrow 1$, $y \leftrightarrow 4$ and $1_{\mathbb{H}} \leftrightarrow 0$. We check the hypotheses of Lemma 4.

- The points $1_{\mathbb{H}}, x, y$ are $0, 1, 4$. They are distinct and all lie on the line D.
- The points x^3, y^3, xy are $3, 12, 5$. They are distinct and lie on the line $\mathcal{M} = -100 + D$, because $103, 112, 105 \in D$.
- $1_{\mathbb{H}}$ is not on $\mathcal{M}$: the point 0 lies on $\mathcal{M}$ if and only if $100 \in D$, which is false.

By Lemma 4, $0 \in x \boxplus y \boxplus z$ but $0 \notin x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$. This proves Theorem 1, so the answer to Question 1 is no. □

Remark 8. The following observation is not used in the proof; it explains why a counterexample cannot be of the most familiar type. Let F be a commutative field and $K^\times$ a subgroup of $F^\times$, and consider Krasner's quotient hyperfield $F/K^\times$ [5], with classes $[X] = XK^\times$ and $[X] \boxplus [Y] = \{[X' + Y'] : X' \in [X],\, Y' \in [Y]\}$. Suppose $0 \in x \boxplus y \boxplus z$ there. Then some $w \in x \boxplus y$ satisfies $0 \in w \boxplus z$, so $w = -z$; hence there are representatives $X \in x$, $Y \in y$ with $X + Y \in -z$, and $Z := -(X + Y)$ is a representative of z with $X + Y + Z = 0$. The identity

$$X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$$

shows that $X^3 + Y^3 + Z^3 + (-XYZ) + (-XYZ) + (-XYZ) = 0$, and $X^3, Y^3, Z^3, -XYZ$ are representatives of $x^3, y^3, z^3, -xyz$; so 0 lies in the six-term sum. Heuristically, this is why the counterexample above uses a projective plane that is not of field origin. We do not prove here that the plane of Lemma 7 is non-Desarguesian, and nothing depends on it.

> **Reviewer 3**
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> Why to mention "non-Desarguesian"?

References

[1] M. Baker and N. Bowler, Matroids over partial hyperstructures, *Adv. Math.* **343** (2019), 821–863.

[2] A. Connes and C. Consani, The hyperring of adèle classes, *J. Number Theory* **131** (2011), 159–194.

[3] M. Hall, Jr., Cyclic projective planes, *Duke Math. J.* **14** (1947), 1079–1090.

[4] M. Krasner, Approximation des corps valués complets de caractéristique $p \neq 0$ par ceux de caractéristique 0, in: *Colloque d'algèbre supérieure (Bruxelles, 1956)*, 1957.

[5] M. Krasner, A class of hyperrings and hyperfields, *Internat. J. Math. Math. Sci.* **6** (1983), 307–311.

[6] O. Viro, Hyperfields for tropical geometry I. Hyperfields and dequantization, preprint, arXiv:1006.3034 (2010).